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Experiment No. 4

Aim:

To design a gas detection system using Arduino Uno, MQ-5 gas sensor, LED, resistor, and connecting wires to detect harmful gases like LPG and methane and provide a visual indication using an LED.

Explanation:

Components and Their Detailed Description

1. Arduino Uno

- **Description**: A microcontroller board based on the ATmega328P, commonly used for embedded systems and DIY electronics projects.

- **Features**:
  - 0 14 digital input/output pins (6 PWM outputs).
  - ○ 6 analog inputs.
  - ○ Operates at 5V with a clock speed of 16 MHz.

- **Function**:
  - 0 Acts as the central control unit, processing sensor data and controlling output devices.
  - ○ Reads input from the MQ-5 sensor and determines if the gas level is dangerous.
  - ○ Controls the LED to turn ON/OFF based on the gas concentration.

2. MQ-5 Gas Sensor

- **Description**: A semiconductor gas sensor used to detect combustible gases like LPG, methane, and natural gas.

- **Features**:
  - ○ Operates on 5V power supply.
  - ○ Provides both digital and analog output.
  - ○ High sensitivity to LPG, propane, and methane.

- **Function**:
  - Detects gas concentration in the surrounding air.
  - Outputs an analog voltage corresponding to the gas level.
  - Sends the signal to the Arduino for further processing.

3. LED Light

- **Description**: A Light Emitting Diode (LED) that glows when an electric current passes through it.

- **Function**:
  - Serves as a visual alert system when a high gas concentration is detected.
  - Controlled by the Arduino using a digital output pin. ○ Helps indicate system status (ON/OFF conditions).

4. Resistor

- **Description**: A passive electrical component that limits current flow.

- **Function**:
  - Used to protect the LED from excess current.
  - Helps stabilize the circuit by providing controlled voltage and current distribution.

5. Connecting Wires

- **Description**: Electrical conductors used to connect different components.

- **Function**:
  - Provide connections between the Arduino, MQ-5 sensor, LED, and power supply.
  - Enable smooth transmission of electrical signals.

Working of the Arduino-Based Gas Detection System

1. **Powering the System**
   - The Arduino Uno and MQ-5 sensor are powered using a **5V power supply**.
   - The sensor starts detecting the gas levels in the surrounding air.

2. **Reading Sensor Data**
   ○ The **MQ-5 sensor outputs an analog voltage** that varies with gas concentration.
   ○ The Arduino reads this signal from **analog input pin A0**.

3. **Processing the Data in Arduino**
   ○ The Arduino compares the sensor reading with a **predefined threshold**.
   ○ If the gas level is **below the threshold**, the LED remains **OFF**.
   ○ If the gas level **exceeds the threshold**, the LED turns **ON** as a warning signal.

4. **Output Indication**
   ○ **No gas detected → LED remains OFF**.
   ○ **High gas concentration → LED turns ON** to alert the user.
   ○ Additional components like a buzzer or display can be added for enhanced alert mechanisms.

Conclusion

This project successfully demonstrates a simple gas leakage detection system using Arduino. The MQ-5 sensor detects gas levels, and an LED provides a visual alert when dangerous concentrations are detected. This system can be further enhanced by integrating a buzzer or a GSM module to send alerts.